

Evaluation Choices Determine Apparent Quantum-Kernel Robustness under Distribution Shift

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Abstract

Quantum-kernel robustness claims depend on how candidates are tuned and selected. We test this dependence in a controlled kernel-swap benchmark spanning four security scenarios, three shift mechanisms, two classifiers and fifteen pipeline realizations per setting. Under same-test OOD-oracle selection and fixed regularization, fidelity kernels appear to improve balanced accuracy over linear and radial-basis-function baselines by up to $+0.037$. When regularization is tuned from training data, selection uses no OOD labels and candidate budgets are matched, the dataset-equal quantum-minus-classical effect becomes -0.005 for support-vector classification and -0.002 for Gaussian-process classification. The primary support-vector result is classical-favoured or conditionally tied in seven of eight scenario-groups. At matched effective rank, classical kernels are at least as accurate in every group. Effective rank and OOD kernel-target alignment are associated with accuracy but favour neither family, while finite-shot estimation distorts exact-statevector effective rank. Evaluation therefore removes the apparent robustness advantage within the low-qubit regimes studied.

Keywords: quantum machine learning; quantum kernels; distribution shift; out-of-distribution robustness; kernel methods; kernel-target alignment; Gaussian processes; malware detection; network intrusion detection

1 Introduction

Quantum kernel methods embed classical data into quantum states and use state overlaps as similarities for otherwise classical learners [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). This separation makes them a clean setting for asking whether quantum-induced feature geometry provides useful inductive bias. It also makes comparison unusually sensitive to evaluation design: the downstream learner can be held fixed, but conclusions still depend on the encoding, bandwidth, regularization, classical reference set, and rule used to select a candidate [Huang et al. \(2021\)](#); [Schuld et al. \(2021\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#).

Recent benchmarks have accordingly shifted the question from whether an isolated quantum model can perform well to whether a claimed gain survives strong and symmetric controls. Broad QML studies find competitive classical baselines once pipelines are aligned [Bowles et al. \(2024\)](#); quantum-kernel benchmarks show substantial sensitivity to feature map and hyperparameter choice [Schnabel and Roth \(2025\)](#); and a concurrent tabular study using nested cross-validation, spectral analysis, and hardware validation finds no significant quantum win across its tested pairs [Kakavand et al. \(2026\)](#). These studies evaluate predominantly identically distributed train–test settings. They do not address a further deployment question: what happens when the target distribution changes and its labels cannot be used to choose the deployed kernel?

Distribution shift makes model selection part of the estimand. Work on domain generalization and real-world shift has shown that robustness gains can disappear when search budgets, hyperparameters, and selection criteria are standardized [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#); [Zhou et al. \(2023\)](#). A selector that inspects the shifted test labels estimates an oracle upper bound, not deployable performance; a family with more candidates has more opportunities to win that selection. Fair quantum–classical evaluation under shift must therefore control

not only the classifier and data split, but also regularization, candidate budget, and label access.

Security detection provides a useful stress test because shift is intrinsic to the application. Malware evolves over time, and experimental bias can overstate detector performance [Anderson and Roth \(2018\)](#); [Pendlebury et al. \(2019\)](#); [Arp et al. \(2022\)](#); network traffic changes with infrastructure and attack campaigns [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#). Quantum classifiers have already been evaluated on intrusion-detection benchmarks, sometimes with reported gains over small classical reference sets on i.i.d. splits [Abreu et al. \(2024, 2025\)](#); [Bellante et al. \(2025\)](#). We use these datasets not to propose a security system, but to expose kernel families to reproducible constructed shifts and to a held-out attack-campaign shift.

The closest robustness-specific concurrent work studies additive input corruption and proposes a spectral phase encoding [Herrero Gómez et al. \(2026\)](#). Our question is complementary: rather than optimize a new encoding, we audit how evaluation choices change a family-level conclusion under constructed and held-out-campaign shifts. The distinguishing axes are no-OOD-label selection, equal candidate budgets, per-configuration train-only regularization, fixed-case inference, and a geometry analysis tied specifically to behaviour after the distribution moves.

We implement a controlled *kernel-swap* benchmark in which preprocessing, splits, classifier, and family-internal selection logic are fixed while only the kernel changes. Four security scenarios from EMBER, UNSW-NB15, and ToN-IoT yield eight scenario-groups across constructed and held-out-campaign shifts. Support-vector and Laplace Gaussian-process classifiers consume the same precomputed Gram matrices. The quantum family contains four fidelity feature-map families with angle-scale sweeps; the classical family extends linear and RBF references with polynomial, Laplacian, and Matérn kernels and symmetric bandwidth freedom. Each configuration’s regularization is tuned from training data alone. The primary deployed candidate is

selected on a disjoint ID-validation split, and the two extended families receive equal candidate budgets. Fifteen pipeline realizations per setting quantify conditional variability; the benchmarks are treated as fixed case studies rather than as random draws from an undefined population.

The comparison is coupled to a mechanistic audit. We measure effective rank, centered kernel-target alignment, concentration, and geometric difference [Roy and Vetterli \(2007\)](#); [Cristianini et al. \(2001\)](#); [Cortes et al. \(2012\)](#); [Huang et al. \(2021\)](#); use partial association, leave-one-dataset-out prediction, and effective-rank matching to restrict interpretation; and perturb fidelity estimates with finite shots to test whether exact statevector geometry survives measurement.

The evaluation choices change the conclusion. With same-test oracle selection, fixed regularization, and customary linear+RBF baselines, fidelity kernels appear to improve OOD balanced accuracy by as much as +0.037. With train-only regularization, no-OOD-label selection, and equal candidate budgets, the dataset-equal quantum-minus-classical delta is -0.005 for SVC and -0.002 for the GP classifier. The primary SVC comparison is classical-favoured or conditionally tied in seven of eight scenario-groups. Effective rank and OOD alignment remain associated with accuracy, but the association is regime-dependent and shared by both families; at matched effective rank, the classical configuration is at least as accurate in every group. Finite-shot estimation inflates effective rank and destabilizes candidate selection.

The contribution is therefore an evaluation result rather than an advantage claim: it identifies which protocol choices create the optimistic quantum-kernel robustness result, supplies a deployable equal-budget comparison under shift, and shows that the observed geometry does not uniquely identify quantum provenance. The scope is deliberately limited to the low-qubit fidelity kernels, datasets, and shift constructions studied here.

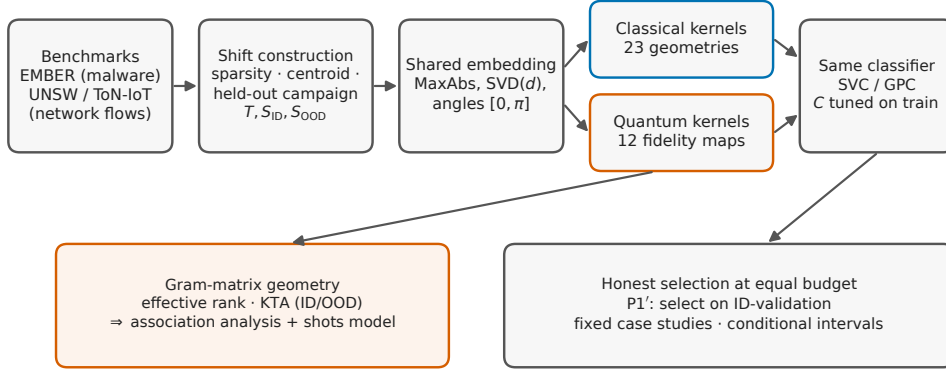


Fig. 1 The controlled kernel-swap protocol. Benchmark pools feed reproducible shift constructions; all kernels operate on a shared train-only embedding, receive per-configuration regularization tuned on the training split, and are consumed by the same precomputed-kernel classifiers. Family-level conclusions are drawn by honest, equal-budget selection on an in-distribution validation split (P1'), summarized as fixed case studies with conditional intervals; the Gram matrices feed the geometry-association analysis and the finite-shot model.

2 Results

2.1 Overview

We report results under the controlled kernel-swap protocol described in Section 4. All comparisons share the same classifier, preprocessing pipeline, and ID/OOD split definitions within a scenario-group; the only varying component is the kernel, and every number below aggregates fifteen pipeline realizations per setting. The argument proceeds in three movements. First, under test-peeking oracle selection and fixed regularization, the fidelity kernels show an apparent OOD advantage over linear and RBF baselines (Section 2.2). Second, we tune each configuration on training data alone, select on an in-distribution validation split disjoint from both test splits, and match the candidate budgets of the extended families. Under this controlled comparison no robust family-level advantage remains; the primary SVC comparison is classical-favoured or tied in seven of eight scenario-groups (Section 2.3). Third, a geometry analysis shows that effective rank and OOD alignment are associated with—but do

not causally determine—performance, that the association is regime-dependent, and that it favours neither family (Section 2.4); a finite-shot analysis then shows the exact geometry does not transfer unchanged to finite measurement (Section 2.5). Throughout, we treat the eight scenario-groups as fixed case studies and report effect sizes with conditional intervals rather than a population p -value, because the settings within a benchmark share data pools and are not exchangeable (Section 4).

2.2 The apparent advantage under oracle selection

We begin with the comparison that dominates the literature: fidelity-based quantum kernels against linear and RBF baselines, with each family’s representative chosen by Best-by-OOD selection—the configuration with the highest balanced accuracy on the OOD test split itself—and the classical kernels left at the default regularization $C = 1$. Under these conventions, on EMBER, the quantum family shows an OOD advantage of +0.037 under SVC and +0.028 under the GP classifier against linear+RBF, shrinking to +0.004/+0.002 once the classical family is enlarged with heavier-tailed kernels. Supplementary Table 2 gives the complete oracle results.

Two features favour this apparent quantum result. Best-by-OOD is an *oracle*: it uses the labels it then reports. Fixing $C = 1$ also leaves classical candidates at a potentially unsuitable regularization point. These values are therefore descriptive upper bounds, not deployable effects.

2.3 A fair comparison removes the advantage

A deployable comparison must tune each configuration’s regularization without looking at the test data, select the deployed configuration without OOD labels, and compare families at the same candidate budget. We implement all three (Section 4). For every kernel candidate, the SVC regularization C is chosen by five-fold cross-validation on the training Gram matrix alone. The deployed configuration is then the one with the best balanced accuracy on an in-distribution *validation* split, held out

from the ID test split (protocol P1’); no OOD label is ever consulted in selection. The primary comparison uses the same candidate budget for both extended families (60 per family in full-coverage groups and 20 in reduced-coverage groups), with the full classical pool retained as a sensitivity. We treat the eight scenario-groups as fixed case studies and report, per group, the quantum-minus-classical OOD delta with a conditional interval over the five split realizations, and no population p -value (Section 4).

Table 1 and Figure 2 give the result: no robust advantage remains against either reference. Against the equal-budget extended classical family, the P1’ delta is classical-favoured or within the conditional uncertainty of zero in seven of the eight scenario-groups under the primary SVC comparison, with a dataset-equal-weighted mean of -0.005 (leave-one-dataset-out range $[-0.006, -0.004]$); under the GP classifier the mean is -0.002 ($[-0.003, +0.000]$). Against the customary linear+RBF baseline—the comparison that produced the $+0.037$ oracle result of Section 2.2—the no-OD-label delta is -0.004 under SVC and $+0.004$ under the GP classifier, with heterogeneous signs across scenario-groups. The single regime that retains a small quantum lean under both equal-budget comparisons is UNSW-NB15 reconnaissance under held-out attack-campaign shift ($+0.005$ for both classifiers); both conditional intervals include zero. The full-pool extended-family sensitivity is similar in direction and scale (dataset-equal means -0.004 for both classifiers), so the conclusion does not depend on giving the classical family its larger pool.

Rank matching does not reveal a residual quantum advantage.

We pair each quantum configuration with the classical configuration of nearest training effective rank, within the same run and dimension, and compare their OOD accuracy. Figure 3 reports the outcome: in all eight scenario-groups the median rank-matched delta is negative (quantum minus classical, between -0.001 and -0.009), and the quantum configuration is the more accurate of the pair in only 22–37% of matches. Within

Table 1 No-OOD-label selection results (eight scenario-groups, fifteen pipeline realizations per setting). Quantum-minus-classical OOD balanced-accuracy delta under P1' (select on ID-validation, tune per-configuration C on train, evaluate on OOD test). The extended-family comparison is the frozen equal-budget analysis: 60 candidates per family in full-coverage groups and 20 in reduced-coverage groups. Its interval is conditional pipeline-realization uncertainty over the five split seeds. The customary linear+RBF comparison is contextual rather than an equal-budget endpoint; the full-pool extended-family sensitivity remains in the machine-readable artifact. No population p -value is reported.

Scenario	Clf.	vs lin./RBF	vs ext. (equal B)	conditional interval
EMBER $m1$	SVC	-0.0095	-0.0058	$[-0.0340, +0.0223]$
	GPC	-0.0050	-0.0009	$[-0.0101, +0.0083]$
EMBER $m2$	SVC	+0.0139	-0.0064	$[-0.0137, +0.0009]$
	GPC	+0.0149	+0.0027	$[-0.0027, +0.0081]$
UNSW-DoS (campaign)	SVC	-0.0043	-0.0033	$[-0.0139, +0.0073]$
	GPC	+0.0041	-0.0057	$[-0.0120, +0.0005]$
UNSW-DoS (constructed)	SVC	-0.0085	-0.0044	$[-0.0119, +0.0031]$
	GPC	-0.0056	-0.0108	$[-0.0177, -0.0039]$
UNSW-Recon (campaign)	SVC	+0.0004	+0.0050	$[-0.0058, +0.0157]$
	GPC	+0.0074	+0.0052	$[-0.0019, +0.0123]$
UNSW-Recon (constructed)	SVC	-0.0076	-0.0089	$[-0.0146, -0.0031]$
	GPC	-0.0083	-0.0075	$[-0.0208, +0.0059]$
ToN-IoT (campaign)	SVC	-0.0172	-0.0148	$[-0.0268, -0.0029]$
	GPC	-0.0046	-0.0090	$[-0.0190, +0.0010]$
ToN-IoT (constructed)	SVC	-0.0016	-0.0011	$[-0.0049, +0.0028]$
	GPC	+0.0313	+0.0109	$[+0.0017, +0.0202]$
Dataset-equal mean, SVC		-0.0043	-0.0050	$[-0.0060, -0.0040]$
Dataset-equal mean, GPC		+0.0043	-0.0019	$[-0.0028, +0.0003]$

the overlap covered by this observational matching, classical kernels are therefore at least as accurate at comparable effective rank.

2.4 Geometry is associated with, but does not determine, robustness

The two families are close because both reach similar kernel geometry, and geometry—not quantumness—is what carries information about behaviour under shift. That information, however, is partial: it is a regime-dependent association, not the single governing law that optimistic accounts (our own earlier report included) proposed.

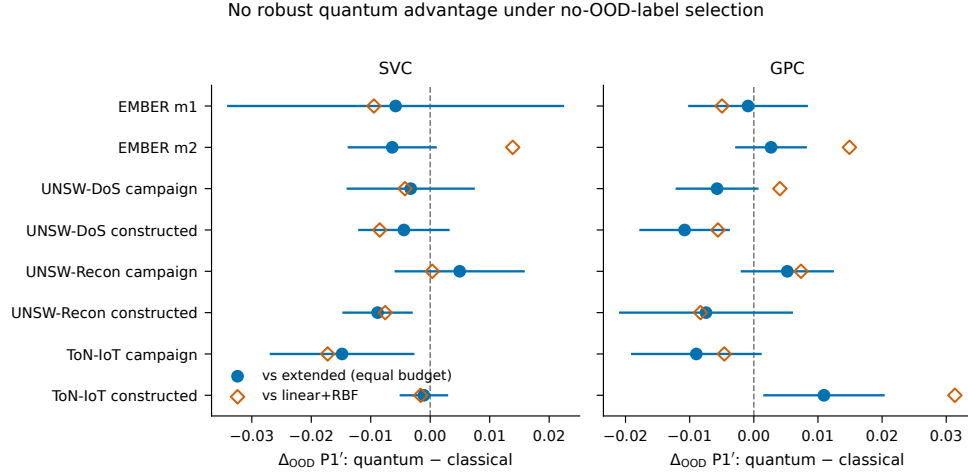


Fig. 2 No robust quantum advantage under no-OOD-label selection. Per-scenario-group P1' OOD deltas (quantum minus classical), for SVC (left) and the GP classifier (right). Filled circles compare against the equal-budget extended classical family with conditional pipeline-realization intervals; open diamonds compare against the customary linear+RBF reference. All eight scenario-groups are shown.

The geometry–OOD link is a regime-dependent association, not a law.

Two descriptors of the training and OOD Gram matrices carry information about which kernels generalize: the effective rank of the training kernel and the centered kernel-target alignment on the OOD split. But their association with OOD accuracy is regime-dependent, not a single mechanism. Within scenario-groups (Figure 4 and Supplementary Table 3), the median Spearman correlation between effective rank and OOD accuracy ranges from +0.80 (EMBER *m2*) down to −0.15 (UNSW-NB15 reconnaissance under the constructed shift); partial correlations that control for family label and length scale track the raw values closely. OOD alignment is more consistently positive (+0.22 to +0.64) but likewise varies by regime. A geometry model fit on three datasets and used to rank the held-out dataset’s kernels achieves only Spearman 0.23–0.40. We therefore describe a genuine but partial association, not a causal mechanism or deployable predictor.

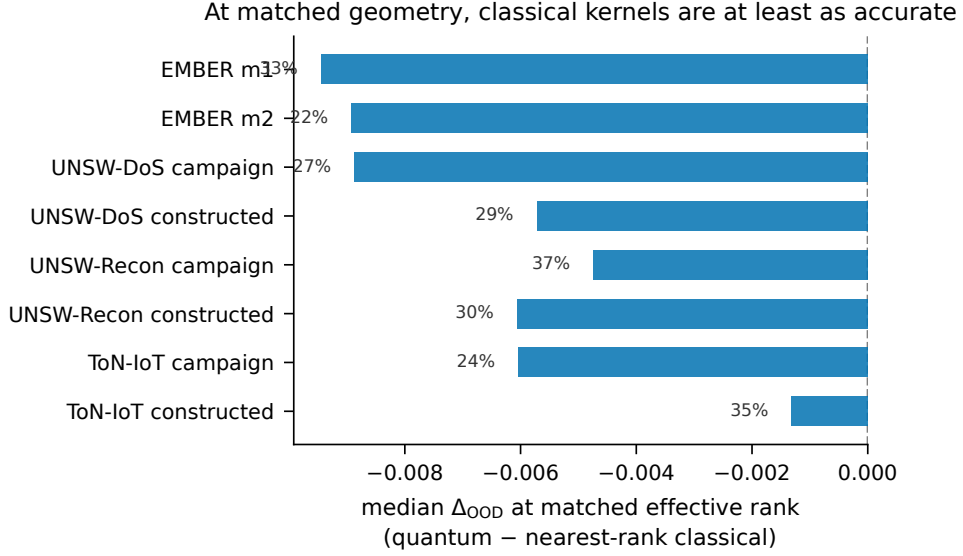


Fig. 3 At matched effective rank, classical kernels are at least as accurate. For each quantum configuration we find the classical configuration of nearest training effective rank (within run and dimension) and take the OOD balanced-accuracy difference; bars are per-scenario-group medians, annotated with the fraction of matched pairs in which the quantum configuration is the more accurate.

2.5 Finite-shot estimation: the geometry is a statevector idealization

All quantum kernels above are computed by exact statevector simulation, which is the right idealization for a geometry study but not what a device would return. To probe how the picture degrades away from that idealization we apply a *finite-shot fidelity-estimation perturbation model*: each fidelity entry is replaced by a binomial estimate from s shots, train blocks are symmetrized, unit-diagonal, clipped to $[0, 1]$, and projected to the positive-semidefinite cone (Section 4). This is not a hardware simulation—no device noise, mitigation, or transpilation is modelled—but it isolates the statistical cost of estimating a fidelity from finite measurements. Table 2 reports the effect on the $q1000$ subset. OOD alignment and OOD accuracy are robust: median KTA_{OOD} deviates by at most 0.0004 from the exact value even at 128 shots, and OOD

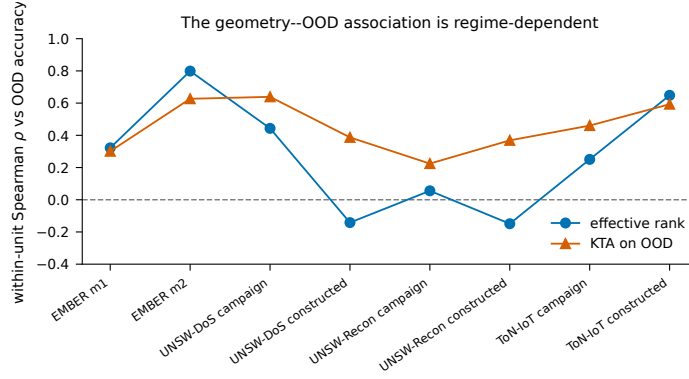


Fig. 4 The geometry–OOD association is regime-dependent. Median within-unit Spearman correlation between each geometry descriptor and OOD balanced accuracy, per scenario-group (SVC). Effective rank orders accuracy strongly in some regimes (EMBER *m2*, ToN-IoT *m2*) and weakly or negatively in others (the UNSW constructed-shift scenarios); OOD alignment is more consistently positive but never uniform. No single descriptor governs all regimes.

balanced accuracy by 0.018–0.033. Effective rank is not: finite sampling inflates it by a median factor of 2.1 at 128 shots, falling to 1.15 only at 8192 shots, because sampling noise adds spurious spectral mass. The within-family configuration selected by ID-validation changes identity under perturbation (agreement 7–21%), but the accuracy cost of that instability is negligible (mean regret ≤ 0.003), so the flatness of the family optimum—already visible in the exact results—persists. The practical reading: the alignment description we can measure transfers to finite shots, but the effective-rank description, and any story that leans on it, is a statevector-exact quantity that would need many shots to reproduce.

2.6 Predictive uncertainty under shift

The Laplace GP probabilities provide a secondary uncertainty check (Supplementary Figure 1). For the honestly selected candidate, predictive entropy rises from ID to OOD for both families in every scenario-group, while expected calibration error changes moderately and comparably. Because neither family has an accuracy advantage, the fidelity kernels show no calibration benefit unavailable to classical kernels. We do not interpret rising entropy alone as evidence of calibration.

Table 2 Finite-shot fidelity-estimation perturbation (EMBER+network $q1000$ subset, SVC). “OOD acc. dev.” is the median absolute deviation of OOD balanced accuracy from the statevector-exact value; “KTA dev.” the same for OOD alignment; “rank ratio” the median ratio of perturbed to exact effective rank; “sel. agree” the fraction of runs whose ID-validation-selected configuration is unchanged; “regret” the mean OOD accuracy lost by deploying the perturbed selection.

Shots	OOD acc. dev.	KTA dev.	rank ratio	sel. agree	regret
128	0.0327	0.0004	2.12	7%	+0.0007
512	0.0273	0.0001	1.56	10%	+0.0017
2048	0.0240	0.0001	1.29	19%	−0.0025
8192	0.0181	0.0000	1.15	21%	−0.0001

3 Discussion

3.1 What is, and is not, a quantum advantage here

The study’s central interpretive move is to replace the question “do quantum kernels beat classical kernels under shift?” with “which properties of a kernel are associated with its behaviour under shift, and does either family have exclusive access to them?”. Read through that lens, the results are clear and negative. The apparent advantage over linear and RBF baselines does not survive the combined correction of oracle selection, fixed regularization, and unequal candidate families. It does not reappear against the equal-budget extended family or at matched effective rank. The geometry descriptors carry information as regime-dependent associations rather than a governing law, and they are shared: a well-tuned heavy-tailed classical kernel occupies the same observed geometric regime as the fidelity maps. On present evidence there is no robust out-of-distribution advantage for fidelity kernels in the low-qubit settings studied, and the only residual—a modest in-distribution separability edge under the probabilistic classifier—is small, classifier-dependent, and itself shrinks under no-OOD-label selection.

From the perspective of the distribution-shift literature, this reinforces a general lesson: conclusions about robustness are only as credible as the protocol used to obtain

them [Quiñonero-Candela et al. \(2008\)](#); [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#). Our results add a QML-specific corollary. The strength of the classical baseline, whether its regularization is tuned, whether model selection may consult the target labels, and the size of each family’s candidate budget are not implementation details: each can move the quantum-versus-classical verdict, and together they turn an apparent $+0.037$ advantage into a null. A demonstration that fixes any of them in the quantum family’s favour, reported alone, would mislead—and much of the optimistic quantum-kernel literature fixes several at once.

3.2 The ID–OOD dissociation is informative

In-distribution separability is the one axis on which the fidelity kernels retain any edge, and even it must be stated carefully. Under the oracle that selects and evaluates on the same in-distribution split, the quantum family separates the classes slightly better than linear+RBF in every EMBER setting; but under honest selection—choosing on a validation split and reporting on a disjoint test split—the edge falls to $+0.003$ under SVC (mixed in sign across groups) and $+0.012$ under the probabilistic classifier, and it does not translate into any out-of-distribution benefit. This ID–OOD dissociation is real but modest, and it cuts against, not for, an operational recommendation: a family that is marginally more separable in-distribution yet no more robust under shift offers nothing to a deployment whose whole difficulty is the shift. The pattern is consistent with the “accuracy on the line” literature [Miller et al. \(2021\)](#), in which ID and OOD accuracy usually move together; here they decouple weakly, and geometry—shared by both families—rather than quantumness marks the axis along which they do.

3.3 The mechanism in context

The geometry association connects several strands of literature. That length scale is decisive for kernel behaviour is known for both families [Shaydulin and Wild \(2022\)](#);

599 [Canatar et al. \(2023\)](#); our contribution is the comparative consequence—once length
600 scale is tuned symmetrically and regularization is tuned at all, the classical fam-
601 ily reaches the same geometric regime, so asymmetric tuning does not merely bias
602 quantum-versus-classical comparisons, it manufactures the apparent advantage. The
603 geometric difference of Huang et al. [Huang et al. \(2021\)](#) does not predict which fam-
604 ily wins a scenario, consistent with its definition as a bound on the *potential* for a
605 separation rather than its sign; a large geometric difference is compatible with equal
606 or worse OOD accuracy, as we observe. That effective rank and OOD alignment are
607 associated with performance is a statement about kernels in general, not about quan-
608 tum kernels—which is precisely why the fidelity maps hold no privileged position: they
609 are one costly route into a geometric regime that a short-length-scale classical kernel
610 reaches directly.

611 The high-rank finding must also be reconciled with two spectral results that super-
612 ficially point the other way. Kübler et al. argue that quantum kernels generalize
613 in-distribution only when the relevant spectral mass is concentrated in few compo-
614 nents [Kübler et al. \(2021\)](#), and Thanasilp et al. show that fidelity kernels concentrate
615 exponentially—toward effectively rank-one, uninformative Gram matrices—as qubit
616 counts grow [Thanasilp et al. \(2024\)](#). There is no contradiction, but there is a genuine
617 tension that our results locate precisely. Both results concern the asymptotic or i.i.d.
618 regime; our effective ranks (roughly 10–150 on training sets of 10^3 – 4×10^3 samples at
619 4–12 qubits) sit far below the concentration regime, so no contradiction arises. Our
620 finite-shot analysis, however, adds an empirical caveat these theoretical results antic-
621 ipate: effective rank measured from finitely estimated fidelities inflates substantially
622 (Section 2.5), so the exact effective ranks we report are a statevector idealization, and
623 any account that leans on effective rank as the operative quantity would need many
624 shots to reproduce even at the modest qubit counts studied here.

We emphasize what the geometry analysis does not deliver. It is a post-hoc association, not a deployable selection rule: choosing a configuration by training-set geometry alone does not reliably identify the better kernel, and OOD alignment requires OOD labels. Because the two families are statistically indistinguishable under honest selection, the more useful open problem is no longer “which family wins” but whether any label-free, train-only signal can select a robust kernel at all—a question that concerns classical and quantum kernels equally.

3.4 Implications for security machine learning practice

From a practical perspective, the paper suggests three takeaways for security ML. The first is substantive: on the evidence here, fidelity-based quantum kernels do not offer a robustness advantage under distribution shift over well-tuned classical kernels, while their pairwise overlap estimation introduces circuit and shot costs absent from direct evaluation of the classical baselines. A median-heuristic Laplacian kernel, especially one with a shortened length scale and its regularization tuned, is a strong and inexpensive robustness baseline, and it matches the fidelity maps under no-OOD-label selection on both modalities. Practitioners weighing a quantum-kernel pipeline for OOD-sensitive security detection should, on this evidence, prefer a tuned classical kernel unless a specific regime demonstrably favours the quantum construction after the same controls.

The second is methodological. Independently of the preferred kernel family, the protocol used here provides a reusable blueprint for robust evaluation in security settings:

1. keep the classifier and preprocessing fixed, and swap only the kernel;
2. define explicit OOD splits tied to interpretable shift mechanisms, including at least one non-constructed shift mechanism where the data permit;

- 691 3. give every family the same freedom—the same candidate budget, symmetric length-
692 scale tuning, and per-configuration regularization tuned on training data alone;
693
694 4. select the deployed configuration without out-of-distribution labels, and keep an
695 in-distribution test split disjoint from the validation split used for selection;
696
697 5. repeat the pipeline, report variability as conditional intervals, and treat correlated
698 benchmarks as fixed case studies rather than pooling them into a single population
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700 p -value.
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702

703 Security ML frequently suffers from comparisons in which model class, tuning effort,
704 and evaluation conditions change simultaneously; the way the verdict of Sections 2.2–
705 2.3 moves from a large apparent advantage to a null as the protocol is made fair is a
706 concrete demonstration of how much the conclusion depends on getting this right.
707

708 The third concerns diagnostics: the geometry descriptors are cheap to compute
709 from Gram matrices that the pipeline already builds, and the alignment-survival anal-
710 ysis can be run post hoc on any deployed kernel system with a labeled drift sample.
711
712 Practitioners should nonetheless resist reading these results as a blanket recommen-
713 dation to replace classical kernels; the responsible recommendation is to test the
714 candidate families under the expected shift regime, using exactly the kind of protocol
715 described here.
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723 3.5 Limitations and threats to validity

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725 Several limitations define the scope of our conclusions.

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728 *Shift operationalization.*

729

730 Our OOD splits cover three reproducible mechanisms—within-class sparsity-tail
731 extremes, train-geometry-tail extremes, and a held-out attack-campaign shift on net-
732 work traffic (an unseen attack campaign plus a capture-partition change, not a
733 timestamped temporal split)—but not the full space of real-world drift processes.
734
735

736

Explicit temporal splits on timestamped malware, family-based drift, and adversarially induced shift remain outside the study’s scope, and our conclusions should be re-tested under them rather than assumed.

Kernel and encoding coverage.

The classical family now spans seven kernels plus a bandwidth sweep, and both families receive symmetric tuning freedom, but neither candidate set is exhaustive: the quantum side is restricted to four commonly used fidelity feature maps at low qubit counts, and trained or task-adapted encodings, projected kernels, and deeper maps are not evaluated. The mechanism gives a concrete prediction for such extensions—what should matter is their Gram-matrix geometry, not their provenance—which future work can falsify. At substantially larger qubit counts, exponential kernel concentration [Thanasilp et al. \(2024\)](#) would erode the alignment-preserving rank the mechanism rewards, so our conclusions should not be extrapolated beyond the low-qubit regime studied here.

Selection over candidate sets.

P1’ (ID-validation selection), P2 (cross-realization OOD-supervised selection), and P3 (same-test OOD oracle) are selection procedures over finite candidate sets. The primary comparison matches candidate budgets, and the full-pool and budget curves provide sensitivities, but residual dependence on candidate definitions and search granularity remains. Fully nested selection over additional independent benchmark pools would strengthen the analysis further.

Simulation, not hardware.

All quantum kernels are computed by exact statevector simulation. The study therefore characterizes the geometry of fidelity kernels, not the behavior of noisy devices:

finite sampling of kernel entries, hardware noise, and error mitigation would all perturb the Gram matrices [Wang et al. \(2021\)](#), and exponential concentration implies that with polynomially many shots the very spectral richness our mechanism rewards becomes expensive to resolve at larger scales [Thanasilp et al. \(2024\)](#); the effect of either on alignment survival is unknown and worth measuring. We make no claim about hardware feasibility, resource advantage, or quantum speedup—the only rigorously established quantum-classical learning separation concerns a specifically constructed problem, not natural data [Liu et al. \(2021\)](#). At the dimensions studied, the fidelity kernels are also classically simulable [Schreiber et al. \(2023\)](#); [Jerbi et al. \(2024\)](#), which is precisely what makes them available as practical kernel constructions today.

Dataset scope and feature caveats.

Four benchmark scenarios across two modalities is a substantial widening relative to single-dataset studies, but security is broader still: dynamic malware analysis, encrypted traffic, and non-security modalities remain untested. Within the network scenarios, the feature sets inherited from the public benchmarks retain port numbers and protocol indicators, which are legitimate but known shortcut-prone features in NIDS benchmarks [Engelen et al. \(2021\)](#); because every kernel receives identical features, this affects absolute accuracies, not the family comparisons. Finally, the mechanism’s effective-rank component is itself regime-dependent (strong on EMBER and ToN-IoT, weak on UNSW-Reconnaissance), a gradation we report rather than smooth over.

3.6 Conservative interpretation

In light of these limitations, the appropriate interpretation of the paper is deliberately conservative. We do not claim universal superiority, complexity-theoretic quantum advantage, or hardware relevance—nor do we claim to have refuted quantum kernels in general. We claim three narrower things, supported by conditional effect estimates

rather than a population p -value: that the apparent out-of-distribution advantage of fidelity kernels over customary baselines disappears when same-test oracle selection and untuned regularization are removed; that under equal candidate budgets and no-OOD-label selection no robust family-level advantage remains against either the customary or the extended classical reference, across eight fixed scenario-groups and two classifiers; and that the geometric descriptors associated with out-of-distribution behaviour occur in both families and do not identify kernel provenance as the source of robustness. This view is consistent with recent calls for careful benchmarking in QML [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#); [Kakavand et al. \(2026\)](#) and is more informative than either an unqualified advantage claim or an unqualified dismissal: it identifies the evaluation choices that generate the optimistic result and shows how the conclusion changes when they are controlled.

3.7 Conclusion

This paper asked why kernel families behave differently under distribution shift, using the quantum-versus-classical comparison as both motivation and stress test. A controlled kernel-swap protocol—classifier, preprocessing, splits, selection logic, and tuning freedom held fixed while only the kernel varies—was instantiated on four security scenarios spanning two modalities and three shift mechanisms, two classifier families including a probabilistic one, and 72 principal settings with fifteen pipeline realizations. Effects are reported as fixed-case summaries with conditional pipeline-realization intervals, not as population significance tests.

The empirical arc has three movements. Under same-test oracle selection and fixed regularization, fidelity-based quantum kernels appear to beat the customary linear and RBF baselines under shift. That result does not survive the controlled comparison: once each configuration is tuned on training data, selection uses no OOD labels, and

875 both extended families search an equal budget of kernel geometries, the dataset-equal-
876 weighted OOD difference is -0.005 under the primary SVC analysis and -0.002 under
877 the GP classifier. The primary comparison is classical-favoured or tied in seven of eight
878 scenario-groups. At matched effective rank the classical kernels are at least as accurate
880 everywhere. Effective rank and OOD alignment are associated with performance in a
881 regime-dependent way, are shared by both families, and, a finite-shot analysis shows,
882 do not transfer unchanged from exact statevectors to finite measurement.

886 The contribution is therefore threefold. Empirically, the study provides a con-
887 trolled, variability-aware comparison of quantum and classical kernels under shift
888 whose central finding is a clean negative obtained under fair conditions: no robust
889 family-level advantage survives budget matching, regularization tuning, and honest
890 selection. Methodologically, it isolates the specific evaluation choices—oracle selection,
891 fixed regularization, mismatched budgets, and pooled significance testing over corre-
892 lated settings—that manufacture apparent quantum-kernel advantages, and it leaves
893 behind a reproducible protocol that removes each. Analytically, it shows that the
894 geometric quantities often invoked to explain such advantages describe both families
895 equally and do not transfer unchanged to finite-shot estimation.

903 The most useful open problem this leaves is not “which family wins” but whether
904 any label-free, train-only signal can select a robust kernel under shift at all; the security
905 community’s label-free drift detectors [Yang et al. \(2021\)](#) and the theory of task–model
906 alignment [Bordelon et al. \(2020\)](#); [Canatar et al. \(2021\)](#); [Ma et al. \(2023\)](#) are natural
907 starting points. It is a question about kernels in general, and answering it would benefit
908 classical and quantum methods alike—which is, in the end, the point: the productive
909 question is not whether one family dominates, but which similarity geometry aligns
910 with the shift regime one actually faces, and how to find it without peeking.

4 Methods	921
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4.1 Problem setting	923
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We study binary security classification—malware versus benign software, and attack	926
versus benign traffic—under distribution shift. Let D_{ID} denote the in-distribution	927
data-generating process and D_{OOD} an out-of-distribution process induced by a con-	928
structed stress test or a held-out attack campaign. Given training data $T \sim D_{\text{ID}}$, we	929
evaluate on an ID test set S_{ID} and an OOD test set S_{OOD} .	930
	931
	932
To isolate the kernel, classifier, preprocessing, optimization protocol, and split	933
design are fixed. Classical and quantum Gram matrices are therefore consumed by the	934
same downstream learner on the same samples.	935
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4.2 Datasets and split construction	940
	941
The four scenarios span static malware (EMBER) Anderson and Roth (2018) and	942
network flows: UNSW-NB15 DoS, UNSW-NB15 Reconnaissance Moustafa and Slay	943
(2015) , and ToN-IoT Scanning Alsaedi et al. (2020) . Every setting has shared T , S_{ID} ,	944
and S_{OOD} samples with identical class balance across kernel families.	945
	946
	947
EMBER uses two label-conditional tail stress tests. The $m1$ split holds out within-	948
class extremes of feature-block sparsity; $m2$ holds out within-class extremes of distance	949
to train-fitted class centroids. Network scenarios use an analogous train-centroid con-	950
struction and a held-out attack-campaign regime. In the latter, training and ID pools	951
contain benign traffic plus non-target attacks, whereas OOD contains benign traffic	952
plus the target campaign; UNSW pools additionally differ in capture partition. These	953
are reproducible stress tests, not timestamped temporal splits or certified covariate	954
shifts. Exact algorithms, feature exclusions, leakage checks, and pool definitions are	955
in the Supplementary Note “Dataset and shift construction.”	956
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967 4.3 Controlled kernel-swap protocol

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969 Figure 1 summarizes the controlled pipeline. Both SVC and a binary Gaussian-process
970 classifier with a logistic-link Laplace approximation consume unit-diagonal precom-
971 puted Gram matrices [Cortes and Vapnik \(1995\)](#); [Rasmussen and Williams \(2006\)](#).
972 Comparisons are always within a classifier. The GP classifier supplies approximate
973 probabilities for uncertainty diagnostics; no post-hoc calibration is applied.
974
975

976 Training-fitted MaxAbs scaling and TruncatedSVD produce shared representations
977 at $d \in \{4, 6, 8, 10, 12\}$. Quantum inputs are then mapped linearly to $[0, \pi]$. Dimension
978 is an explicit candidate axis.
979
980

983 4.4 Kernel families

984
985 The customary classical reference contains linear and RBF kernels. The extended
986 reference adds degree-2 and degree-3 polynomial, Laplacian, and Matérn-3/2 and
987 5/2 kernels. RBF, Laplacian, and Matérn bandwidths sweep factors $\{0.1, 0.3, 1, 3, 10\}$
988 around training-only scale or median-distance heuristics. The quantum family uses
989 fidelity
990

$$991 K_Q(x, x') = |\langle \phi_Q(x), \phi_Q(x') \rangle|^2$$

992 for two ZZ depths, PauliXZ, and Z feature maps [Havlíček et al. \(2019\)](#); [Schuld and](#)
993 [Killoran \(2019\)](#). Every map sweeps angle scales $\{0.5, 1, 2\}$. Classical bandwidth and
994 quantum angle-scale freedom are symmetric in purpose [Shaydulin and Wild \(2022\)](#);
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1003 “Controlled kernel-swap protocol.”
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1006 4.5 Selection, candidate budgets, and inference

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1008 For each geometry, SVC regularization $C \in \{0.01, 0.1, 1, 10, 100\}$ is selected by five-
1009 fold stratified cross-validation on the training Gram matrix. The primary P1’ protocol
1010 divides the ID holdout into disjoint validation and test halves. Within each family,
1011
1012

the candidate with the highest ID-validation balanced accuracy is deployed on ID-test and OOD-test; OOD labels never enter this choice. Same-test OOD-oracle and cross-realization OOD-supervised protocols are descriptive sensitivities.

Best-of-family selection favours a larger pool. The primary extended-family endpoint therefore uses 60 candidates per family where coverage is complete and 20 where it is reduced, generated by 5,000 classical-pool subsamples under three schemes. The full 115-candidate classical pool is secondary.

Every configuration has fifteen pipeline realizations (five split seeds by three model seeds). Two EMBER and six network scenario-groups are crossed with three master seeds and $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$ regimes (1000, 500, 500), (2000, 1000, 1000), and (4000, 1800, 1800). Because scenario-groups share source pools and are not exchangeable, they are fixed case studies. Per group we average over the five split-seed clusters and report a conditional interval; across groups we use dataset-equal means and leave-one-dataset-out ranges. We attach no population p -value to eight fixed, correlated cases. The complete grid is in Supplementary Table 1 and resampling details in the Supplementary Note “Controlled kernel-swap protocol.”

4.6 Metrics and geometry analysis

OOD balanced accuracy is primary. Family effects are quantum minus classical after selection. ID balanced accuracy and ID-to-OOD degradation are secondary. For the GP classifier we also record log loss, Brier score, 15-bin expected calibration error [Guo et al. \(2017\)](#), and predictive entropy.

From the classifier Gram matrices we compute effective rank [Roy and Vetterli \(2007\)](#), centered kernel-target alignment [Cristianini et al. \(2001\)](#); [Cortes et al. \(2012\)](#), entry concentration, cross-split similarity, and regularized geometric difference [Huang et al. \(2021\)](#). We relate geometry to OOD accuracy within scenario-groups, use partial association to control family and length scale, test portability by leaving one dataset

out, and pair quantum candidates with nearest-effective-rank classical candidates within run and dimension. These analyses are associational. Formulas and diagnostic definitions appear in the Supplementary Note “Metrics and geometry descriptors.”

4.7 Finite-shot perturbation

Exact fidelity entries are replaced by binomial estimates at $s \in \{128, 512, 2048, 8192\}$ shots. Training blocks are symmetrized, clipped, set to unit diagonal, and projected to the positive-semidefinite cone. The model isolates sampling error, not device noise. Projection diagnostics and interpretation are in the Supplementary Note “Finite-shot fidelity-estimation model.”

4.8 Use of generative AI

OpenAI Codex was used during manuscript revision to assist with language editing, code inspection, and consistency checking. The authors reviewed and take responsibility for all analyses, claims, citations, and text; the system is not an author.

Declarations

Data availability. This study uses the public EMBER [Anderson and Roth \(2018\)](#), UNSW-NB15 [Moustafa and Slay \(2015\)](#), and ToN-IoT [Alsaedi et al. \(2020\)](#) benchmarks, which remain subject to their original access conditions. Derived split definitions, shift constructions, experiment manifests, aggregated results, and summary tables are available in the Zenodo repository at <https://doi.org/10.5281/zenodo.19147649>. Before submission, this DOI will be verified against the exact archived release used to build the manuscript.

Code availability. Code for constructing the splits, running the kernel experiments, reproducing the confirmatory analysis, and regenerating every table and figure is available at https://github.com/roberto-fernandez-barrios/kernel_shift_framework and in

the versioned Zenodo archive above. The platform-independent Python 3.12 environment is pinned in `environment.lock.yml`; the code is released under BSD-3-Clause with no restriction on non-academic use.

Competing interests. The authors declare that they have no competing interests.

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Authors' contributions. RFB: conceptualization, methodology, software, data curation, formal analysis, visualization, investigation, writing—original draft, writing—review and editing. IPL: supervision, validation, writing—review and editing. AGS: validation, project administration, writing—review and editing. PGB: supervision, resources, writing—review and editing. All authors read and approved the final manuscript.

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